# Parallel Adversarial Domain Adaptation for Cross-Dataset Hyperspectral Image Classification

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***Abstract*—With the advancement of spectral imaging technology, hyperspectral image (HSI) resources have been rapidly expanding, making cross-dataset HSI classification a critical technique and an inevitable trend for large-scale Earth observation applications. However, most existing approaches transfer from one HSI to another in a supervised way, which limits their ability to leverage multisource HSI information. Therefore, this article proposes an unsupervised cross-dataset HSI classification method based on parallel adversarial domain adaptation (PADA), which learns and integrates task-relevant and domain-invariant knowledge from multisource HSIs to classify a target HSI. Specifically, a source-and-target shared information mining module is designed to mine transferable knowledge from each source HSI. This module employs parallel adversarial learning between a spectral–spatial feature extractor and a task-relevant controller with a domain-invariant discriminator to learn task-relevant and domain-invariant features, thereby mitigating domain shift. Moreover, a source-to-target transferability learning module is proposed to evaluate the cross-dataset transferability of knowledge, which computes a domain correlation score to evaluate interdomain correlation and guide adaptive knowledge transfer, effectively suppressing redundant information and reducing negative transfer caused by low-correlated sources. Finally, a multisource collaborative classification module is developed to accomplish cross-dataset knowledge transfer, which designs a correlation-aware fusion strategy to produce the final classification result by integrating information from each source, ensuring balanced and robust decision-making. Extensive experiments conducted under challenging multisource cross-dataset settings validate the classification performance and the domain extensibility of the proposed method, demonstrating its potential for large-scale Earth observation applications.**

***Index Terms*—Cross-dataset classification, domain adaptation, hyperspectral image (HSI).**

## I. Introduction

HYPERSPECTRAL image (HSI), which integrates wide-coverage spatial information with high-resolution spectral information, has overcome the limitations of traditional remote sensing images and demonstrated irreplaceable value in various fields, such as precision agriculture and mineral exploration. HSI classification, which leverages spatial and spectral information to produce a pixel-level land-cover map, effectively facilitates the transition of remote sensing tasks from qualitative decision-making to quantitative analysis and has become a frontier research direction and a key focus in the field of remote sensing.

Classical HSI classification research is mainly based on manual spectral and spatial feature extraction techniques. Spectral feature extraction is primarily concerned with analyzing the continuous spectral curves of pixels and achieves effective expression of spectral information through dimensionality reduction and filtering techniques, such as principal component analysis [1], independent component analysis [2], linear discriminant analysis, and wavelet transform [3]. Spatial feature extraction uses the spatial correlation of pixel neighborhoods to mine the spatial contextual characteristics through texture analysis methods, such as morphological profiles [4], gray-level co-occurrence matrices, and Gabor filters. Classical methods laid the foundation for HSI classification and conducted effective explorations for the application of HSIs. Despite their effectiveness, these traditional methods heavily rely on manually engineered features, limiting their ability to capture complex nonlinear relationships and to handle challenges such as spectral variability and noise inherent in HSI data.

Deep learning approaches can automatically extract hierarchical features from low-level features to high-level semantic representations [5] and have been introduced to address the issues of classical HSI classification methods. Some works improved convolutional neural networks (CNNs) for HSI classification, significantly enhancing classification performance through end-to-end spectral–spatial feature learning [6]. Roy et al. [7] proposed a HybridSN method, which integrates 3-D and 2-D CNNs to jointly extract spatial–spectral features from stacked spectral bands. Wang et al. [8] introduced an end-to-end cubic-CNN, which efficiently captures spatial and spatial–spectral features by performing convolutional operations on different directional planes of feature cubes. Zheng et al. [9] proposed a method that combines 2-D and 3-D convolutions with covariance pooling to extract and fuse spatial–spectral features. Li et al. [10] proposed the CDFS-CASCL framework, which extends CNN-based spectral–spatial feature extractors to cross-domain few-shot

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scenarios through cross-modal alignment and supervised contrastive learning. It performs parallel spatial–spectral feature extraction with equal-weight fusion, enhancing the discriminability and generalization of learned representations under limited training samples.

Transformer has also been introduced in HSI classification to mine spectral correlations by modeling global context with self-attention mechanisms [11]. Wu et al. [12] proposed the spectral–spatial window attention Transformer, which combines fused-MBConv, cross-window attention, and interactive feature-enhanced modules to capture both local and long-range dependencies in HSIs, achieving state-of-the-art classification accuracy and robustness. Ahmad et al. [13] proposed DiffFormer, a differential spatial–spectral Transformer that incorporates differential multihead self-attention, SwiGLU activation, and patch-based spectral–spatial tokenization to enhance subtle spectral–spatial variations and mitigate redundancy. Yang et al. [14] proposed an adaptive coupling Transformer network, a hybrid network combining CNN and Transformer branches with an adaptive response fusion module and cosine similarity-based mutual learning to capture both local and global spectral–spatial representations.

For single-dataset or distribution-consistent HSI classification tasks, deep learning-based methods have significantly improved classification performance through their powerful feature extraction capabilities. However, for large-scale cross-dataset scenarios, some challenges still need to be addressed. On the one hand, most learning-based methods heavily rely on large-scale labeled samples, which are often unavailable in remote sensing applications with continuously emerging HSIs. On the other hand, most learning-based methods assume that the training and testing samples share the same distribution, which is challenged by the cross-dataset distribution shifts caused by variations in geography, sensors, and acquisition conditions. Therefore, it is imperative to develop HSI classification methods that enable effective cross-dataset generalization under limited sample conditions.

Transfer learning facilitates target HSI classification by leveraging knowledge learned from a known source HSI, which provides a feasible solution for cross-dataset HSI classification under limited sample conditions [17]. Zhang et al. [18] proposed a masked diffusion framework with a highly efficient Transformer backbone, which effectively captures transferable representations across scenes for cross-dataset HSI classification. Li et al. [19] combined few-shot learning with self-supervised learning, leveraging both natural images and limited HSI samples to transfer spatial meta-knowledge and extract discriminative spectral–spatial representations. The aforementioned methods provide a possible solution for cross-dataset HSI classification. However, when the transfer scenario extends to a completely unknown and unlabeled target HSI, achieving effective model generalization for the HSI with a new scenario becomes a more challenging, yet more meaningful research problem.

Unsupervised domain adaptation (UDA), a branch of transfer learning, has been proposed to address the aforementioned challenge. Wang et al. [20] used graph neural networks with joint coral to enhance the smoothness of the features while aligning the marginal and conditional distributions. Wang et al. [21] designed a domain-adversarial broad adaptation network to reduce both marginal and second-order statistical differences. Makkar et al. [22] proposed a multitask adversarial framework to align target features with the source through unsupervised or semi-supervised strategies. Ma et al. [23] introduced a variational adversarial network that enforces class boundary constraints during feature alignment. By bridging the distribution gap between the source HSI and target HSI, UDA-based methods enhance the model's generalization ability on the target HSI, providing a solid technical foundation for cross-dataset HSI classification. Although single-source UDA has made progress in reducing the distribution shift between source and target domains, relying on a single source often fails to achieve robust cross-dataset generalization. Single-source approaches are constrained by large source–target discrepancies, limited class coverage, and the risk of negative transfer, which restricts their generalization capability. These limitations highlight the necessity of multisource domain adaptation. By integrating information from multiple heterogeneous HSIs, multisource methods can obtain more diverse and comprehensive feature representations, which enhances the robustness and generalization of models across domains.

However, multisource adaptation is not a straightforward extension of single-source methods. In cross-dataset HSI classification, different source domains often differ in spectral distributions, spatial structures, sensor characteristics, and data quality. Directly aggregating all sources without distinction may introduce redundant or conflicting information, which can result in negative transfer and degraded target performance. The relevance between each source and the target domain also varies. Uniform aggregation may allow low-relevance sources to dominate and further impair classification performance. Therefore, multisource adaptation requires carefully designed strategies that extract domain-invariant, yet task-relevant features and adaptively weight each source according to its relevance to the target domain to achieve effective and robust cross-dataset transfer.

Inspired by the aforementioned methods, this article proposes a multisource cross-dataset HSI classification method based on parallel adversarial domain adaptation (PADA) to transfer domain-invariant and task-related knowledge from multisource labeled HSIs to an unlabeled target HSI. Specifically, the proposed method designs a source-and-target shared information mining module to learn what to transfer, a source-to-target transferability learning module to quantify how much to transfer, and a multisource collaborative classification module to optimize how to transfer. By parallel adversarial learning, domain-invariant and task-related features are captured from multiple source HSIs without any labeled target sample. With domain consensus evaluation and correlation-aware fusion, the target classification map is generated by adaptively integrating information from multiple source HSIs. Through supervised learning on multisource HSIs, the proposed method obtains generalized task-relevant knowledge, facilitating continuous unsupervised classification of the target HSI.

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In summary, the main contributions of this article are as follows.

1) To leverage multisource HSI information, this article proposes an unsupervised multisource cross-dataset HSI classification method based on PADA, which transfers domain-invariant and task-related knowledge from multisource HSIs to a target HSI, and makes multisource collaborative classification with a correlation-aware fusion strategy.
2) To extract transferable knowledge from each source HSI, this article designs a source-and-target shared information mining module, which explores local structure-aware spatial features with TriConvBlock and global context-aware spectral features with spectral-oriented Transformer, and adversarial learns with a task-relevant controller to guide the extraction of discriminative task-specific features, and a domain-invariant discriminator to capture shared domain-invariant features.
3) To evaluate the knowledge transferability of each source HSI, this article designs a source-to-target transferability learning module, which utilizes a domain consensus evaluator to compute interdomain correlation and adaptively guide knowledge transfer from multisource HSIs.

The remainder of this article is organized as follows. Section II introduces UDA techniques and adversarial learning. Section III provides a detailed description of the proposed cross-dataset HSI classification method. Section IV presents the experiments with analysis, including datasets, experimental settings, comparative experiments, and ablation studies. Finally, Section V concludes the article.

## II. Related Work

This section reviews recent advances relevant to our work, focusing on UDA and adversarial learning. These techniques have shown great promise in improving cross-dataset HSI classification by aligning distributions and enhancing feature learning.

### A. Unsupervised Domain Adaptation

In HSI classification, the acquisition of labeled samples is labor-intensive and time-consuming. UDA, a pivotal subfield of transfer learning, transfers a classification model from a labeled source HSI to an unlabeled target HSI, thereby addressing the challenge of insufficient labeled samples [16]. UDA-based HSI classification methods can promote generalization when limited labeled samples are available.

Existing UDA-based HSI classification methods can be broadly categorized into two types: feature alignment-based methods and adversarial training-based methods. The fundamental principle of feature alignment-based approaches is to learn domain-invariant representations by minimizing the discrepancy between feature distributions [24]. Prominent alignment techniques include maximum mean discrepancy, correlation alignment, and graph optimal transport, which reduce domain shifts by minimizing the statistical distance between the source and target domains. For instance, Wang et al. [25] proposed a pairing domain discrimination network, which aligns source and target domains by distinguishing whether paired samples come from the same domain and enhances feature learning through multilevel mutual information maximization. Dong et al. [26] introduced a contrast-constrained domain adaptation strategy, which enhances feature distribution alignment by maximizing the mutual information between corresponding categories from both domains. Although these methods achieve some level of alignment, they often lack robustness when confronted with complex distribution shifts.

To overcome these limitations, adversarial training-based approaches have emerged as a breakthrough in UDA and have been widely applied in tasks such as image classification. These methods leverage adversarial learning frameworks, such as generative adversarial networks (GANs), to drive feature extractors toward learning domain-invariant representations, thereby promoting alignment between source and target domains. For example, Cui et al. [27] proposed a category semantic-guided adversarial UDA method that constructs a category memory bank via visual–semantic alignment and enhances discriminative learning through bi-classifier adversarial training. Wang et al. [28] proposed a bidirectional classifier adversarial network, which enhances feature distribution alignment by maximizing and minimizing the decision discrepancy between two classifiers, facilitating knowledge transfer across domains. Fang et al. [29] proposed an adversarial UDA framework for HSI classification that integrates masked self-distillation and bi-classifier adversarial training to enhance class separability and exploit target-domain semantics.

Furthermore, with the continuous advancement of deep learning, sophisticated models that integrate multiple techniques are increasingly being applied to UDA for HSI classification. For instance, Shen et al. [30] proposed a hierarchical prototype-aligned graph neural network that uses multiscale graph convolution and prototype alignment to reduce domain shift and improve cross-scene adaptation. Similarly, Feng et al. [31] proposed S4DL, a shift-sensitive spatial–spectral disentangling framework that explicitly models spectral and spatial shifts to achieve robust UDA across diverse HSI scenes.

In summary, UDA offers a promising solution to the challenge of label scarcity in HSI classification by enabling knowledge transfer from labeled source to unlabeled target domains. However, most existing UDA methods focus on a single-source scenario, which limits their ability to exploit the diversity and complementarity among multiple source scenarios. To address this issue, we propose a UDA-based multisource HSI classification framework that is designed to fully capture and integrate heterogeneous information across multiple sources, thereby enhancing domain generalization performance.

### B. Adversarial Learning

Adversarial learning, especially since the introduction of GANs in 2014 [32], has emerged as a powerful paradigm in computer vision tasks such as image generation, style transfer, object detection, and classification. At its core, the GAN

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consists of a generator and a discriminator engaged in a two-player game: the generator aims to synthesize realistic data samples, while the discriminator seeks to distinguish between real and generated data. Through adversarial training, the generator learns to produce increasingly realistic outputs, enabling effective feature alignment and data distribution modeling.

This adversarial mechanism has shown substantial promise in HSI analysis. Beyond conventional data augmentation, adversarial learning has been leveraged to address key challenges in HSI, including the scarcity of labeled samples, severe class imbalance, and domain discrepancy. Qin et al. [33] proposed a distance-constrained GAN that incorporates latent and output distance constraints to stabilize training and improve feature discrimination. Liang et al. [36] introduced a spectral–spatial attention GAN that leverages unlabeled data and employs a mean-minimization loss to alleviate mode collapse. More recently, Chen et al. [34] designed a spectral–spatial adversarial multidomain synthesis network that uses multiscale mutual information regularization to enforce semantic consistency across domains, thereby improving generalization in cross-dataset classification. Feng et al. [35] further advanced adversarial domain adaptation for HSI classification by introducing class-aligned and class-balancing strategies to address domain shift and class imbalance. However, despite these advances, most existing GAN-based methods rely on pairwise adversarial structures, which often fail to capture complex distribution discrepancies across multiple source domains and the target domain. These approaches typically conduct untargeted domain alignment without explicit guidance from task semantics, limiting their ability to extract representations that are both domain-invariant and task-discriminative. Moreover, training instability and poor generalization remain persistent challenges.

To address these shortcomings, we propose a source-and-target shared information mining module based on a novel triangular adversarial structure, where each source domain forms an adversarial cycle composed of a feature extractor, a task-relevant controller, and a domain-invariant discriminator. This design enables dual-level supervision, feature-level alignment, and task-level discrimination, offering explicit task guidance during domain adaptation. Unlike conventional pairwise GANs, our approach transforms domain alignment from an untargeted process into a task-oriented adversarial adaptation strategy, effectively bridging domain gaps while preserving class semantics.

## III. Methodology

The proposed method consists of three functional modules: a source-and-target shared information mining module to learn what to transfer, a source-to-target transferability learning module to quantify how much to transfer, and a multisource collaborative classification module to optimize how to transfer. This section will introduce the problem formulation, three modules, and the model training process, respectively.

### A. Problem Formulation

The objective of the proposed unsupervised cross-dataset HSI classification is to adapt a classification model trained on $N$ labeled source HSIs to an unlabeled target HSI, achieving unsupervised classification of the target HSI by leveraging multisource HSI information. Specifically, given $N$ labeled source HSIs and one unlabeled target HSI, the sample set of the $i$th source HSI can be represented as

$$\left\{\mathbf{X}_i^s, \mathbf{Y}_i^s\right\} = \left\{\left(\mathbf{x}_{i,j}^s, \mathbf{y}_{i,j}^s\right) \mid j = 1, 2, \ldots, M_i^s\right\} \tag{1}$$

where the superscript $s$ designates samples or sets derived from the source HSI, $\mathbf{x}_{i,j}^s$ denotes the $j$th sample from the $i$th source HSI, $\mathbf{y}_{i,j}^s \in \{1, 2, \ldots, K\}$ is its corresponding label vector, $M_i^s$ represents the number of samples in the $i$th source HSI, and $i = 1, 2, \ldots, N$. Similarly, the sample set of the target HSI is denoted as

$$\mathbf{X}^t = \left\{\mathbf{x}_k^t \mid k = 1, 2, \ldots, M^t\right\} \tag{2}$$

where the superscript $t$ designates samples or sets derived from the target HSI, $\mathbf{x}_k^t$ represents the $k$th sample from the target HSI, and $M^t$ is the total number of samples in the target HSI.

The proposed method leverages task-related knowledge learned from multiple labeled HSIs to perform unsupervised classification on the target HSI. Given the supervision information $\{\mathbf{X}_i^s, \mathbf{Y}_i^s\}$ from $N$ known HSIs, we train the proposed PADA network with embedding function $\mathcal{F}_{\text{PADA}}(\cdot)$ to interpret an unknown HSI $\mathbf{X}^t$. Specifically, for each target sample $\mathbf{x}_k^t$, the predicted label is obtained by

$$\hat{\mathbf{y}}_k^t = \mathcal{F}_{\text{PADA}}\left(\mathbf{x}_k^t\right). \tag{3}$$

The overall architecture of the proposed PADA framework is illustrated in Fig. 1. First, the source-and-target shared information mining module learns transferable representations and derives initial predictions. It extracts task-related and domain-invariant spectral–spatial features for each source sample $\mathbf{x}_{i,j}^s$ and each target sample $\mathbf{x}_k^t$ with a spectral–spatial extractor and refines the features by parallel adversarial learning with a task-relevant controller and a domain-invariant discriminator. Second, the source-to-target transferability learning module evaluates transferability and guides adaptive knowledge transfer. It estimates the domain correlation score for each source sample $\mathbf{x}_{i,j}^s$ and each target sample $\mathbf{x}_k^t$ with a domain consensus evaluator to produce cross-domain transferability. Finally, the multisource collaborative classification module integrates the initial predictions from all source branches and produces the final classification decision. It designs a correlation-aware fusion strategy to make decisions for each target sample $\mathbf{x}_k^t$ guided by the transfer weights.

### B. Source-and-Target Shared Information Mining

The source-and-target shared information mining module aims to address the question of "what to transfer." This module consists of $N$ parallel shared information mining branches, one for each source HSI. Each branch forms an adversarial cycle among the spectral–spatial feature extractor ($G_i$), a task-relevant controller ($C_i$), and a domain-invariant discriminator ($D_i$). This adversarial structure simultaneously imposes both discriminative constraints at the feature level and task-related supervision at the prediction level, providing dual guidance for subsequent multisource knowledge fusion and thereby

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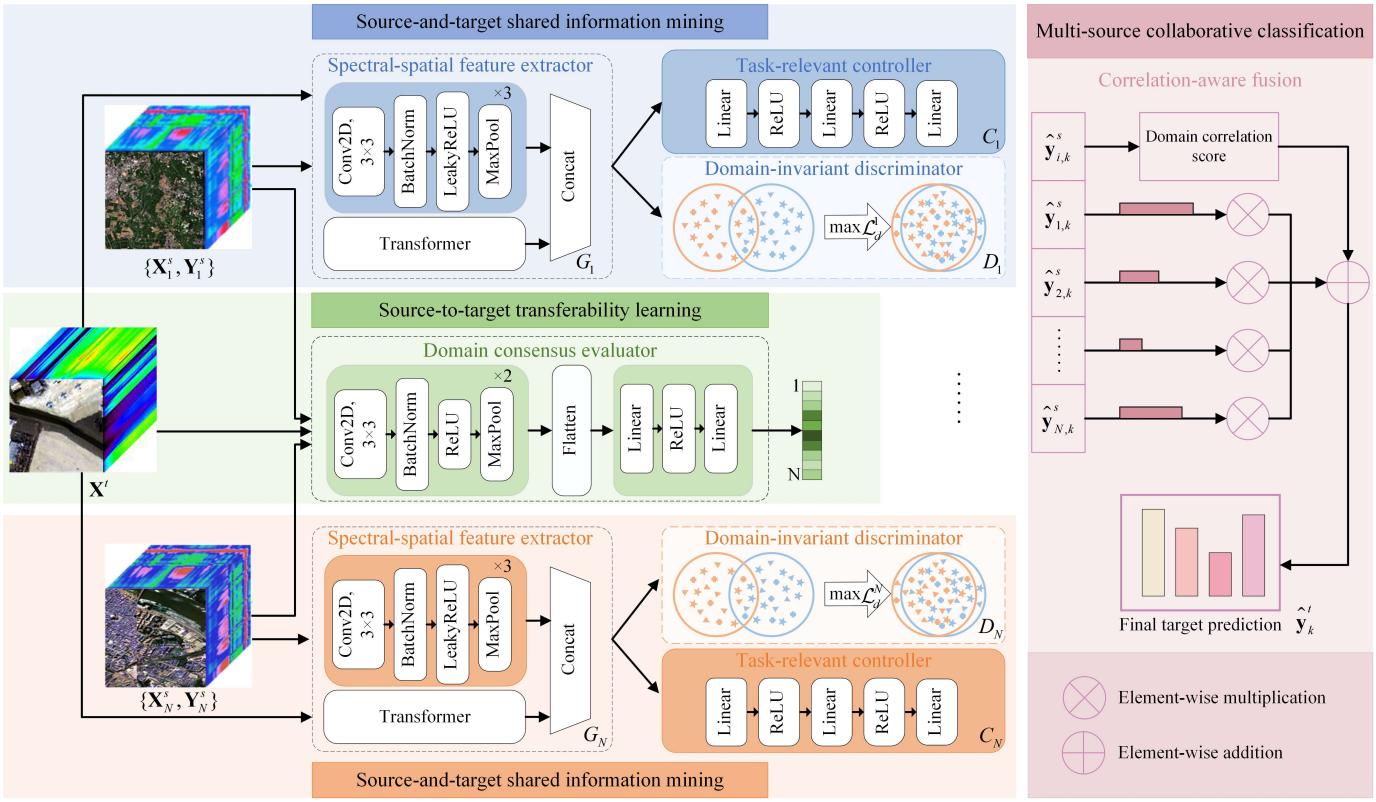



Fig. 1. Overall architecture of the proposed PADA framework. The framework consists of three core functional modules. The source-and-target shared information mining module extracts domain-invariant and task-related features while generating initial predictions. The source-to-target transferability learning module computes the domain correlation score between the target HSI and each source HSI to derive adaptive transfer weights. The multisource collaborative classification module integrates source-specific predictions using the learned transfer weights to produce the final classification result.

enhancing the representation capacity for cross-dataset transfer classification.

The spectral–spatial feature extractor $G_i$ employs a spectral–spatial joint feature extraction structure that independently models spatial and spectral features before integrating them at the feature level, ensuring a comprehensive representation of the multidimensional characteristics inherent in HSIs. The local structure-aware spatial feature extractor adopts the TriConvBlock, which consists of three stacked convolutional layers to hierarchically capture local spatial features. Owing to the strong local correlations in hyperspectral spatial patterns, each convolutional block uses convolutional operations to extract local spatial information, enhanced by a Batchnorm layer and LeakyReLU activation [37] for improved stability. Pooling layers are also incorporated to reduce computational complexity, and the resulting spatial features are flattened into a one-dimensional vector for subsequent fusion with spectral features.

The global context-aware spectral feature extractor leverages a spectral-oriented Transformer architecture [38] to model global spectral relationships and long-range dependencies along the spectral dimension, enhancing spectral representations, as shown in Fig. 2. Unlike traditional CNNs constrained by local receptive fields, Transformers can use global attention mechanisms to address nonlocal correlations effectively. To better accommodate the spectral structure of HSIs, we propose an improved spectral-dimensional patch embedding mechanism. Instead of spatial partitioning in standard Transformers, this method segments features exclusively along the spectral dimension while preserving spatial integrity. Specifically, a 2-D convolution operation is applied along the spectral dimension to generate a patch sequence, with a convolution kernel size of (patch-size,1) and a stride of (patch-size,1). A learnable classification token and positional encoding are then added to the sequence, where the positional encoding parameters are initialized using a normal distribution. The encoded sequence is subsequently processed through multiple Transformer layers, which leverage multihead self-attention mechanisms and feed-forward networks to model long-range spectral dependencies. To provide a clear overview of the proposed spectral–spatial feature extractor, Table I summarizes

TABLE I

ARCHITECTURE AND KEY HYPERPARAMETERS FOR TRICONVBLOCK AND SPECTRAL TRANSFORMER

| Module | Input | Output | Kernel | Depth | Dim | Heads |
|---|---|---|---|---|---|---|
| TriConvBlock-1 | $(C, 25, 25)$ | $(64, 12, 12)$ | $3 \times 3$ | 1 | 64 | – |
| TriConvBlock-2 | $(64, 12, 12)$ | $(128, 6, 6)$ | $3 \times 3$ | 1 | 128 | – |
| TriConvBlock-3 | $(128, 6, 6)$ | $(256, 3, 3)$ | $3 \times 3$ | 1 | 256 | – |
| Spectral-PatchEmbed | $(C, 25, 25)$ | $(C/6, 3750)$ | $6 \times 1$ | 1 | 3750 | – |
| Spectral-Transformer | $(N, 3750)$ | $(N, 3750)$ | – | 2 | 3750 | 6 |
| Spectral-FC | $((N+1) \times 3750)$ | $(100)$ | – | 1 | 100 | – |

Notes: $C$ is the spectral dimension. Spatial size $25 \times 25$. Number of spectral tokens $N = C/s$: $C = 102 \Rightarrow N = 17$ (Pavia); $C = 144 \Rightarrow N = 24$ (Houston). Embedding dim $= 25 \times 25 \times 6 = 3750$.

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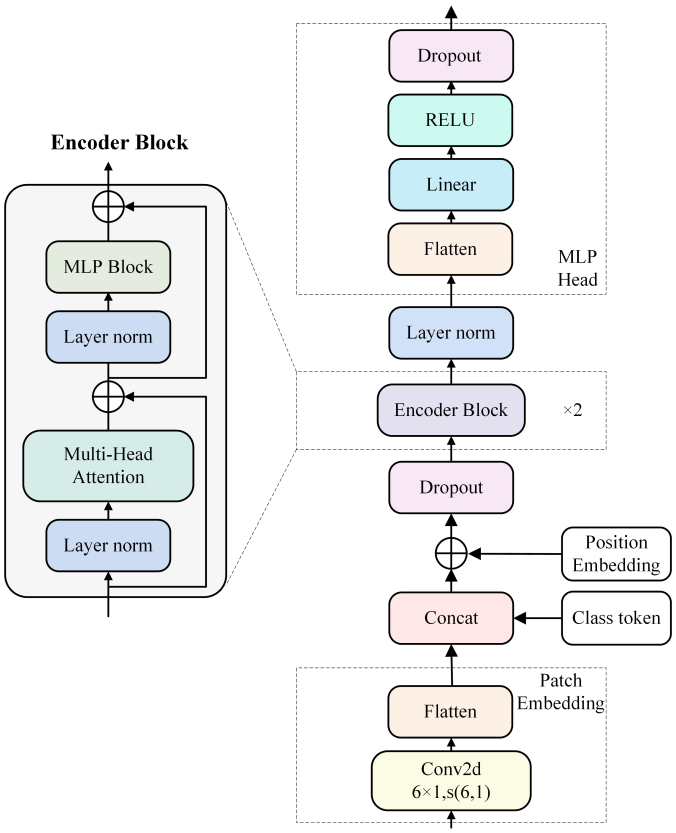



Fig. 2. Global context-aware spectral feature extractor. It extracts features along the spectral dimension using an improved spectral-dimension patch embedding mechanism and leverages a multihead self-attention mechanism to model long-range spectral dependencies.

the architecture and hyperparameters of the TriConvBlock and spectral Transformer.

Finally, feature-level concatenation generates a spectral–spatial joint representation that preserves the multidimensional characteristics of HSIs. Given a source sample $\mathbf{x}_{i,j}^s$, the spectral–spatial feature extractor $G_i$ outputs the spectral–spatial representations

$$\mathbf{z}_{i,j}^s = G_i\left(\mathbf{x}_{i,j}^s\right) \tag{4}$$

and similarly, for a target sample $\mathbf{x}_k^t$, we obtain

$$\mathbf{z}_{i,k}^t = G_i\left(\mathbf{x}_k^t\right). \tag{5}$$

The joint features are then passed through the task-relevant controller $C_i$, which is implemented as a compact three-layer fully connected neural network. It maps the joint features to a category probability distribution, providing discriminative feature supervision under source-domain guidance. Given the extracted joint features $\mathbf{z}_{i,j}^s$, the predicted label vector is computed by

$$\hat{\mathbf{y}}_{i,j}^s = C_i\left(\mathbf{z}_{i,j}^s\right). \tag{6}$$

This progressive dimensionality reduction not only preserves nonlinear modeling capacity but also mitigates overfitting and improves classification performance.

The domain-invariant discriminator $D_i$ enhances domain adaptability via adversarial learning by encouraging the spectral–spatial feature extractor to extract domain-invariant features, thereby reducing domain shift. The module consists of three fully connected layers followed by a Sigmoid activation function. During training, the discriminator aims to accurately distinguish whether the features originate from the source or target HSI, while the feature extractor optimizes its representations to confuse the discriminator, thus promoting domain invariance.

Formally, for the features $\mathbf{z}_{i,j}^s$ from the $i$th source HSI and $\mathbf{z}_{i,k}^t$ from the target HSI, the domain-invariant discriminator predicts the domain labels as

$$\hat{d}_{i,j}^s = D_i\left(\mathbf{z}_{i,j}^s\right) \tag{7}$$

$$\hat{d}_{i,k}^t = D_i\left(\mathbf{z}_{i,k}^t\right). \tag{8}$$

The controller guides the feature extractor to preserve task-relevant semantics, while the adversarial discriminator encourages the suppression of domain-specific variations. Through their joint optimization, the feature extractor learns to produce representations that are simultaneously discriminative and domain-invariant, thereby achieving effective feature separation.

### C. Source-to-Target Transferability Learning

The source-to-target transferability learning module, based on a domain consensus evaluator, addresses the question of "how much to transfer." It quantitatively measures the correlation between the target HSI and each source HSI, and assigns adaptive transfer weights that guide the fusion of multisource knowledge. Serving as a bridge between the source and target shared information mining module and the multisource collaborative classification module, this module plays a critical role in determining the direction and degree of knowledge transfer.

Specifically, the domain consensus evaluator employs a dual-layer convolutional block structure for feature extraction. Each block consists of a 2-D convolutional layer, BatchNorm, ReLU activation, and MaxPooling, which together capture multiscale feature information. The resulting feature maps are then projected into an $N$-dimensional space via two fully connected layers, where $N$ is the number of source HSIs, producing the raw domain correlation score vector. Formally, given a source sample $\mathbf{x}_{i,j}^s$ from the $i$th source HSI, the domain consensus evaluator outputs a raw domain correlation score vector as

$$\tilde{\mathbf{q}}_{i,j}^s = E\left(\mathbf{x}_{i,j}^s\right) \tag{9}$$

where $E$ represents the domain consensus evaluator and $\tilde{\mathbf{q}}_{i,j}^s \in \mathbb{R}^N$. Similarly, for a target sample $\mathbf{x}_k^t$, the evaluator outputs

$$\tilde{\mathbf{q}}_k^t = E\left(\mathbf{x}_k^t\right) \tag{10}$$

where $\tilde{\mathbf{q}}_k^t \in \mathbb{R}^N$ indicates the raw domain correlation score between the target sample and all $N$ source HSIs.

To convert these raw scores into normalized transfer weights suitable for multisource knowledge fusion, we apply the Softmax function along the source dimension

$$\hat{q}_{i,j}^{s,(m)} = \frac{\exp\left(\tilde{q}_{i,j}^{s,(m)}\right)}{\sum_{l=1}^{N}\exp\left(\tilde{q}_{i,j}^{s,(l)}\right)}, \quad m = 1, \ldots, N \tag{11}$$

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and

$$\hat{q}_k^{t,(m)} = \frac{\exp\left(\tilde{q}_k^{t,(m)}\right)}{\sum_{l=1}^{N} \exp\left(\tilde{q}_k^{t,(l)}\right)}, \quad m = 1, \ldots, N. \tag{12}$$

The resulting normalized domain correlation vectors are given by

$$\hat{\mathbf{q}}_{i,j}^{s} = \left[\hat{q}_{i,j}^{s,(1)}, \ldots, \hat{q}_{i,j}^{s,(N)}\right]^{\top} \in [0,1]^N \tag{13}$$

and

$$\hat{\mathbf{q}}_{k}^{t} = \left[\hat{q}_{k}^{t,(1)}, \ldots, \hat{q}_{k}^{t,(N)}\right]^{\top} \in [0,1]^N \tag{14}$$

where each element represents the relative contribution of the corresponding source HSI, with larger values indicating stronger cross-domain alignment.

Through this two-step process, the source-to-target transferability learning module provides a dynamic, interpretable, and data-driven estimation of source-to-target transferability, effectively attenuating the influence of low-correlated sources while emphasizing highly relevant sources for robust multisource domain adaptation.

### D. Multisource Collaborative Classification

The multisource collaborative classification module is designed to address the final stage of cross-dataset decision-making by integrating knowledge from multiple source HSIs in a correlation-aware manner. Specifically, it performs correlation-aware fusion of the prediction outputs from all $N$ task-relevant controllers, leveraging the domain correlation score computed by the source-to-target transferability learning module as dynamic fusion coefficients. This adaptive fusion mechanism ensures that source HSIs with higher semantic and distributional correlation to the target sample contribute more significantly to the final prediction.

Compared to conventional voting or uniform averaging schemes, correlation-aware fusion explicitly incorporates inter-domain alignment information, enabling more informed and selective knowledge aggregation. This design improves classification accuracy and enhances the efficiency of multisource knowledge transfer.

Formally, the final prediction for a target sample $\mathbf{x}_k^t$ is computed as

$$\hat{\mathbf{y}}_k^t = \sum_{i=1}^{N} \hat{q}_k^{t,(i)} \cdot \hat{\mathbf{y}}_{i,k}^t \tag{15}$$

where $\hat{\mathbf{y}}_k^t$ denotes the aggregated label prediction vector for the target sample. $\hat{q}_k^{t,(i)}$ represents the adaptation weight for the $i$th source HSI, and $\hat{\mathbf{y}}_{i,k}^t$ is the predicted label vector for the target sample generated by the $i$th parallel shared information mining branch. The final category of the target sample $\mathbf{x}_k^t$ is determined by selecting the class corresponding to the maximum value in $\hat{\mathbf{y}}_k^t$.

By dynamically weighting source predictions based on transferability, this module effectively filters out noisy or irrelevant knowledge and enhances the reliability of the final classification decision under the multisource domain adaptation setting. In addition, the correlation-aware fusion scheme provides an explicit mechanism to alleviate distributional divergence among multiple sources. Since each source-specific branch generates predictions independently, the adaptive weights act as a gate to filter inconsistent or low-correlated sources. When a source domain exhibits large semantic or distributional divergence from the target, its correlation score will be small, thereby reducing its contribution to the final decision. This design not only mitigates conflicts between heterogeneous sources but also suppresses negative transfer by prioritizing knowledge from well-aligned sources.

### E. Network Training

The training process of the PADA framework involves two main stages.

*1) Training of the Source-and-Target Shared Information Mining Module:* The goal of this stage is to learn joint feature representations that are both domain-invariant and task-related. We train $N$ parallel shared information mining branches, where each branch consists of a spectral–spatial feature extractor $G_i$, a task-relevant controller $C_i$, and a domain-invariant discriminator $D_i$. The procedure is as follows.

Training the task-relevant controller $C_i$: For the $i$th adversarial subnetwork, we first input the sample sets from both the $i$th source HSI and the target HSI into the corresponding spectral–spatial feature extractor $G_i$, thereby obtaining spectral–spatial joint representations $\mathbf{z}_{i,j}^s$ for source samples and $\mathbf{z}_{i,k}^t$ for target samples, respectively. The features extracted from the source samples are then passed into the task-relevant controller $C_i$, which produces the predicted category probability vectors $\hat{\mathbf{y}}_{i,j}^s$. To supervise this mapping process, a standard cross-entropy loss is employed, which compares the predicted label distribution against the corresponding ground-truth label $\mathbf{y}_{i,j}^s$. Then the loss function is defined as

$$\mathcal{L}_c^i = \frac{1}{M_i^s} \sum_{j=1}^{M_i^s} \mathcal{H}_{\text{cross}}\left(\hat{\mathbf{y}}_{i,j}^s, \mathbf{y}_{i,j}^s\right) \tag{16}$$

where $\mathcal{H}_{\text{cross}}$ denotes the cross-entropy loss function and $M_i^s$ is the number of samples in the $i$th source HSI. By optimizing $\mathcal{L}_c^i$, the task-relevant controller $C_i$ is updated to drive it to accurately predict labels based on the extracted features. At the same time, the cross-entropy loss $\mathcal{L}_c^i$ based on the source HSI label constrains the spectral–spatial feature extractor to generate discriminative representations.

Training the domain-invariant discriminator $D_i$. To enforce domain invariance, the hyperspectral features $\mathbf{z}_{i,j}^s$ from the $i$th source HSI and $\mathbf{z}_{i,k}^t$ from the target HSI are utilized to train the domain-invariant discriminator. Specifically, for the $i$th source HSI features $\mathbf{z}_{i,j}^s$, a real domain label $d_{i,j}^s = 1$ is assigned, indicating source domain origin. Similarly, for the target HSI features $\mathbf{z}_{i,k}^t$, a real domain label $d_{i,k}^t = 0$ is assigned. The discriminator $D_i$ then predicts whether the features originate from the source or target domain and is trained using the binary cross-entropy loss

$$\mathcal{L}_d^i = \frac{1}{M_i^s} \sum_{j=1}^{M_i^s} \mathcal{H}_{\text{cross}}\left(\hat{d}_{i,j}^s, d_{i,j}^s\right) + \frac{1}{M^t} \sum_{k=1}^{M^t} \mathcal{H}_{\text{cross}}\left(\hat{d}_{i,k}^t, d_{i,k}^t\right) \tag{17}$$

where $\hat{d}_{i,j}^s$ and $\hat{d}_{i,k}^t$ denote the predicted domain labels for source and target features, respectively, and $M^t$ is the number of unlabeled target samples. By minimizing $\mathcal{L}_d^i$, the

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domain-invariant discriminator $D_i$ is optimized to effectively distinguish whether the features come from the source HSIs or the target HSI.

Adversarial optimization of the spectral–spatial feature extractor $G_i$. Based on the feedback from the $i$th task-relevant controller $C_i$ and the $i$th domain-invariant discriminator $D_i$, the $i$th feature extractor $G_i$ is trained to extract task-related and domain-invariant features. The loss function of the spectral–spatial feature extractor combines the classification constraint and the domain adversarial constraint, and is formulated as follows:

$$\mathcal{L}_g^i(t) = \alpha(t)\,\mathcal{L}_c^i - \beta(t)\,\mathcal{L}_d^i \tag{18}$$

where $\alpha(t)$ and $\beta(t)$ are dynamically adjusted positive weighting parameters. The positive term $\alpha(t)\mathcal{L}_c^i$ strengthens feature discriminability under the supervision of the task-relevant controller, while the negative term $-\beta(t)\mathcal{L}_d^i$ encourages the features to confuse the domain-invariant discriminator.

Motivated by the potential limitations of fixed weighting, we propose a dynamic weighting mechanism based on the historical loss values. This approach allows the weights of the classification supervision and the domain alignment constraint to be adaptively adjusted during training, eliminating the need for manual tuning or grid search. At the beginning of training, the exponential moving averages (EMAs) are initialized as $\mathrm{EMA}_c^i(0) = 0$ and $\mathrm{EMA}_d^i(0) = 0$. During training, the classification loss and domain adversarial loss are smoothed using EMA to reduce batch-level fluctuations

$$\mathrm{EMA}_c(t) = \gamma \cdot \mathrm{EMA}_c(t-1) + (1-\gamma)\cdot \mathcal{L}_c^i(t) \tag{19}$$

$$\mathrm{EMA}_d(t) = \gamma \cdot \mathrm{EMA}_d(t-1) + (1-\gamma)\cdot \mathcal{L}_d^i(t) \tag{20}$$

and the adaptive weighting parameters are then computed as

$$\alpha(t) = \mathrm{clip}\left(\frac{\alpha_0}{\frac{\mathrm{EMA}_d(t)}{\mathrm{EMA}_c(t)+\epsilon}}, \alpha_{\min}, \alpha_{\max}\right) \tag{21}$$

$$\beta(t) = \mathrm{clip}\left(\beta_0 \cdot \frac{\mathrm{EMA}_d(t)}{\mathrm{EMA}_c(t)+\epsilon}, \beta_{\min}, \beta_{\max}\right) \tag{22}$$

where clip$(\cdot)$ constrains the values within a stable range, $\epsilon$ is a small constant to prevent division by zero, and $\gamma$ controls the smoothing effect. Intuitively, when the domain adversarial loss is relatively large compared to the classification loss, the adaptive mechanism increases $\alpha(t)$ to emphasize feature discriminability, while $\beta(t)$ is adjusted to maintain domain alignment; conversely, when the classification loss dominates, $\beta(t)$ is increased to strengthen domain invariance. This dynamic mechanism ensures that the tradeoff between feature discriminability and domain invariance is adaptively balanced throughout training, avoiding over-reliance on manually set hyperparameters and effectively mitigating the potential risk of target domain label leakage. Consequently, the spectral–spatial feature extractor can robustly learn discriminative and domain-invariant representations simultaneously, which is crucial for cross-domain generalization.

This design forms an adversarial cycle of "enhanced discriminability-enhanced domain invariance." Specifically, the task-relevant controller requires the features to be separable across categories, the domain-invariant discriminator requires the features to be indistinguishable across domains, and the feature extractor seeks the Pareto optimality between the two, and finally generates double-excellent features that both retain the ability to distinguish categories and eliminate domain-specific noise. Compared with the pairwise adversarial model, the triangular architecture upgrades feature learning from untargeted domain alignment to task-oriented domain adaptation through bidirectional adversarial supervision, fundamentally improving the transferability and robustness of cross-domain classification.

*2) Training of the Source-to-Target Transferability Learning Module:* This stage aims to build an adaptive relationship between the source HSIs and the target HSI. During the training process, the sample sets of all $N$ source HSIs are used to jointly train the source-to-target transferability learning module. First, a supervisory signal is designed to assign a domain correlation score vector to each source-domain sample in the form of one-hot encoding, thereby forcing the module to learn the explicit mapping between the input sample and its associated source HSI. Specifically, for the sample $\mathbf{x}_{i,j}^s$ of the $i$th source HSI, a true domain correlation score vector is assigned as $\mathbf{q}_{i,j}^s = (q_1, q_2, \ldots, q_N)$ with the $i$th component set to 1 and all other components set to 0, indicating that the sample is most similar to those from the $i$th source HSI. Then, after the sample is processed by the source-to-target transferability learning module, the predicted domain correlation score vector is output

$$\hat{\mathbf{q}}_{i,j}^s = E\left(\mathbf{x}_{i,j}^s\right). \tag{23}$$

The loss function of the source-to-target transferability learning module is defined as

$$\mathcal{L}_e = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{M_i^s}\sum_{j=1}^{M_i^s}\mathcal{H}_{\mathrm{cross}}\left(\hat{\mathbf{q}}_{i,j}^s, \mathbf{q}_{i,j}^s\right) \tag{24}$$

where $M_i^s$ denotes the number of samples in the $i$th source HSI. By optimizing $\mathcal{L}_e$, the source-to-target transferability learning module is driven to accurately capture the correlation relationship between samples and each source HSI, thereby providing an accurate weighted basis for the subsequent multisource collaborative classification module.

## IV. Experiments

This section presents experiments with the corresponding analysis based on four publicly available HSIs. All experiments were conducted on a workstation with an Intel Core i9-10900X processor (3.70 GHz), an NVIDIA TITAN RTX GPU, and 24 GB of RAM, and implemented using Python 3.11 under the PyTorch framework.[1]

### A. Experimental Dataset

To evaluate the effectiveness of the proposed PADA method, experiments are conducted on four HSI datasets: Houston 2013 [39], Houston 2018 [40], Pavia Center, and Pavia University.

1) *Houston 2013 (H13):* H13 was captured by the CASI-1500 sensor in Houston in 2013. This dataset contains 15

[1]PyTorch: https://pytorch.org/

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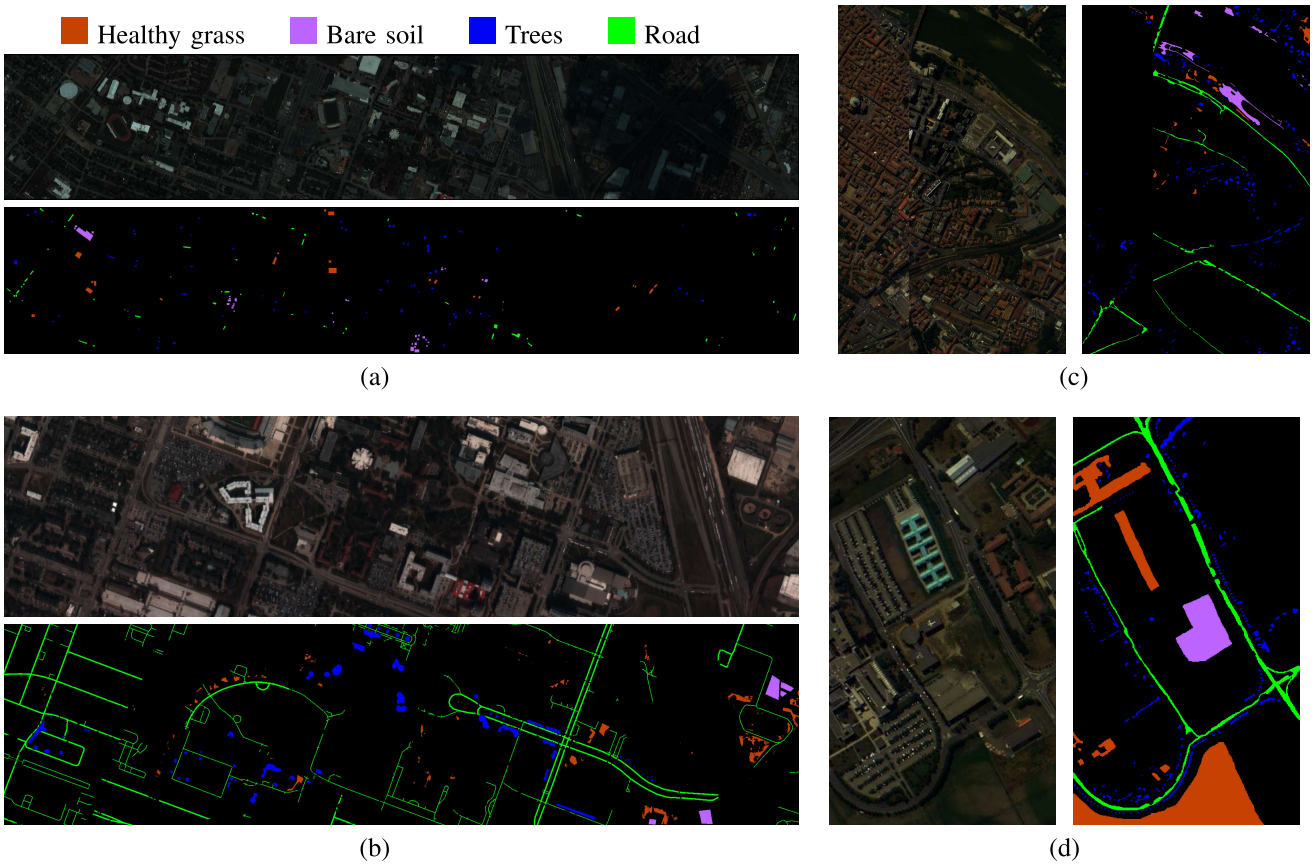



Fig. 3. False-color images and ground-truth maps of different HSIs. (a) Houston 2013. (b) Houston 2018. (c) Pavia Centre. (d) Pavia University.

classes with a spectral range of 0.38–1.05 $\mu$m, 144 spectral bands, and a spatial size of 349 $\times$ 1905. The false-color and ground-truth maps are shown in Fig. 3(a).

2) *Houston 2018 (H18):* H18 was also captured by the CASI-1500 sensor in Houston in 2018. This dataset contains 20 classes with a spectral range of 0.38–1.05 $\mu$m, 48 spectral bands, and a spatial size of 601 $\times$ 2384. The false-color and ground-truth maps are shown in Fig. 3(b).

3) *Pavia Centre (PC):* PC was photographed by the ROSIS-03 sensor in 2003 in the Pavia region of northern Italy. The dataset contains 9 classes, with a spectral range of 0.43–0.86 $\mu$m, 102 spectral bands, and a spatial size of 1096 $\times$ 715. The false-color and ground-truth maps are shown in Fig. 3(c).

4) *Pavia University (PU):* PU was acquired by the ROSIS sensor in 2003 from the University of Pavia in northern Italy. The dataset contains 9 classes with a spectral range of 0.43–0.86 $\mu$m, 103 spectral bands, and a spatial size of 610 $\times$ 340. The false-color and ground-truth maps are shown in Fig. 3(d).

To comprehensively evaluate the generalization capability and robustness of the proposed method in the cross-dataset HSI classification task, we utilize the datasets introduced above and conduct a detailed analysis of their class distributions and acquisition characteristics. To ensure consistency and comparability in cross-domain classification, we focus on the common classes present in all datasets, including healthy grass, bare soil, trees, and roads.

### B. Experimental Setup

Given the significant differences among these datasets in terms of sensor type, acquisition time, geographical coverage, and spectral characteristics, we further design the experimental protocol based on sensor heterogeneity to increase the difficulty and realism of the adaptation setting.

Specifically, inspired by the protocol design in [44], we adopt a cross-sensor partitioning strategy to simulate challenging and realistic scenarios in remote sensing applications. In the first group, H13 and H18 are selected as the source domains, and either PC or PU is used as the target domain. To construct a representative source domain, we include all labeled samples from four selected classes in H13. For H18, we sample 9799, 4516, 10 000, and 10 000 samples from its four selected categories, respectively. All available labeled samples in the target domain are used for testing. In the second group, PC and PU are used as sources, and H13 or H18 as the target. Specifically, we sample 3090, 6584, 6000, and 4000 samples from four selected categories in PC, and 5000, 5029, 3000, and 6631 samples from PU. As in the first group, all labeled samples from the target domain are retained for testing. This grouping strategy introduces greater domain shift in terms of spectral response, spatial resolution, and acquisition mechanisms, thereby constructing a more challenging cross-sensor adaptation scenario that better reflects real-world remote sensing conditions. It also enables a more rigorous assessment of the proposed method's capability to handle domain discrepancies under complex, heterogeneous settings.

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To further eliminate discrepancies in spectral dimensionality across datasets and ensure consistent model inputs, we perform spectral alignment preprocessing within each experimental group. In the first group, the spectral configuration of H13 is used as the reference. The H18 data, which has fewer bands, is extended by duplicating its bands, while PC and PU are resampled based on their original spectral range, and any missing bands are zero-padded. In the second group, PC serves as the spectral reference, and the other datasets are spectrally cropped and resampled to obtain a uniform 102-band input. This preprocessing step standardizes the input dimensionality across all domains, reducing the impact of sensor-induced variation and enabling more reliable evaluation of domain adaptation performance. Spectral dimensionality reduction or band selection has been widely studied to retain informative bands while reducing computational cost and preserving physical interpretability [46]. In this work, the preprocessing strategy is primarily designed to standardize spectral dimensionality across heterogeneous sensors, thereby facilitating stable feature extraction and enabling fair cross-domain comparison. Instead of introducing new algorithmic mechanisms, this standardized alignment ensures comparability and preserves global spectral structures while maintaining physical interpretability at the band level.

For performance evaluation, we adopt overall accuracy (OA, %), average accuracy (AA, %), and the Kappa coefficient ($\kappa$) as objective metrics to assess the proposed method against the comparison approaches. To alleviate the bias introduced by random sampling, multiple independent runs are conducted, and the averaged results are reported. For training, we employ the Adam optimizer with an exponentially decaying learning rate schedule. The initial learning rate is set to $1 \times 10^{-}$ with a decay factor of 0.9, and the batch size is fixed to 128.

Regarding the training budget and convergence criterion, different strategies are adopted for the two core modules. For the source-and-target shared information mining module, an early stopping strategy is employed based on the dynamic behavior of the feature extractor loss. Specifically, let $L_t$ denote the feature extractor loss at the $t$th epoch and $L_{t-1}$ the loss of the previous epoch, and define the change as

$$\Delta L_t = |L_t - L_{t-1}|. \tag{25}$$

If $\Delta L_t$ remains below a threshold $\delta = 10^{-3}$ for $p = 15$ consecutive epochs, the model is regarded as converged, and training is stopped early. The threshold $\delta$ is determined based on the magnitude of the loss and empirical observations to filter out fluctuations caused by individual batches, while the patience parameter $p$ ensures that short-term randomness does not interfere with the overall convergence judgment. The maximum number of training epochs for this module is set to 100. For the source-to-target transferability learning module, the number of training epochs is determined according to the stabilization of the error convergence curve and is fixed at 50 epochs.

For the adaptive weighting mechanism of the feature extractor loss, the hyperparameters are set as follows. The initial scaling coefficients are $\alpha_0 = 1.0$ and $\beta_0 = 1.0$. To maintain stability, the weighting factors are clipped into constrained intervals, with $\alpha_{\min} = 0.1$, $\alpha_{\max} = 10.0$, $\beta_{\min} = 0.1$, and $\beta_{\max} = 10.0$. The EMA smoothing factor is fixed to $\gamma = 0.9$ to balance short-term fluctuations and long-term trends, and a small constant $\epsilon = 10^{-6}$ is introduced to avoid division by zero. These values are determined empirically to provide stable training behavior across different datasets without extensive hyperparameter tuning.

### C. Comparison Results

To validate the effectiveness, the proposed method is compared with six state-of-the-art methods, including SSFTT [15], SSTN [41], AMS-M2ESL [42], SDEnet [43], MDGT [44], and CACL [45].

1) *SSFTT:* Shallow spectral and spatial features are extracted by 3-D and 2-D convolution, respectively, and feature transformation is performed using a Gaussian-weighted tagger, which is input to the Transformer encoder for feature learning and determining the sample labels by a linear layer.

2) *SSTN:* To overcome the limitations of convolutional kernels, the SSTN combines spatial attention and spectral correlation modules. A decomposed architecture search framework is designed to optimize hierarchical operations and module ordering, enabling efficient and stable architecture search.

3) *AMS-M2ESL:* AMS-M2ESL designs an adaptive mask-based intrablock sampling module to sample spatial relations oriented to the center spectral vector. It also proposes a dual-channel distance covariance representation module to uniformly construct spectral–spatial features and mine both linear and nonlinear spectral relationships. In addition, it leverages the Riemannian geometry of the SPD manifold for high-level feature learning and performs efficient Euclidean subspace projection via an approximate matrix square root layer.

4) *SDEnet:* The SDEnet uses generative adversarial learning, trained in the source domain and tested in the target domain. Extended domains are generated via an encoder–randomizer–decoder architecture, using spatial and spectral randomization to generate information and applying morphological knowledge as domain-invariant information. The discriminator learns category-level domain-invariant representations through supervised contrast learning and optimizes the generator to promote the separation of source and extended domain-like samples.

5) *MDGTnet:* The MDGTnet consists of a classifier and two branches. The intradomain uniqueness extraction branch captures the internal attributes of each source domain, while the interdomain commonality extraction branch learns shared features across different source domains. By removing the effects of the imaging environment through multilayer fusion, the network obtains distinctive and discriminative category features.

6) *CACL:* CACL is designed for cross-scene HSI classification under UDA. It separately performs domain-level and class-level distribution alignment. Domain-level alignment is achieved via adversarial training between the feature extractor and the domain discriminator. For class-level alignment, spectral–spatial prototypes of both source and target domains

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TABLE II
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON DIFFERENT TDS

| SD-TD | Class | SSFTT | SSTN | AMS-M2ESL | SDEnet | MDGTnet | CACL | PADA |
|---|---|---|---|---|---|---|---|---|
| H13+H18-PU | 1 | 48.40±30.48 | 0.00±0.00 | 40.60±30.61 | 59.56±33.45 | 50.78±11.47 | **71.42±25.65** | 68.37±12.61 |
| H13+H18-PU | 2 | **77.00±28.16** | 62.20±42.78 | 40.00±33.11 | 27.26±33.80 | 38.49±12.53 | 38.52±35.33 | 64.22±23.05 |
| H13+H18-PU | 3 | 12.40±13.86 | 78.80±27.31 | 16.20±20.47 | 23.17±20.25 | **92.05±4.15** | 16.81±32.88 | 68.55±14.78 |
| H13+H18-PU | 4 | 76.60±18.44 | 19.20±31.94 | 55.40±40.31 | 91.74±10.60 | 87.38±18.47 | 86.60±22.68 | **100.00±0.00** |
| H13+H18-PU | AA(%) | 53.60±5.37** | 39.20±7.33** | 38.00±14.46** | 50.04±5.68** | 67.17±5.76* | 53.34±3.47** | **76.15±5.24** |
| H13+H18-PU | OA(%) | 54.86±13.65 | 20.33±4.31** | 41.20±19.20* | 57.75±14.31* | 59.99±6.68** | 60.47±11.83 | **73.13±3.58** |
| H13+H18-PU | $\kappa$(×100) | 35.80±11.80* | 8.00±4.36** | 0.18±0.18** | 36.17±12.61** | 44.83±7.20** | 43.66±11.44* | **61.52±4.16** |
| H13+H18-PC | 1 | 84.40±25.14 | 0.40±0.55 | 93.00±8.43 | **97.93±1.47** | 92.78±4.06 | 31.07±39.62 | 69.66±8.57 |
| H13+H18-PC | 2 | 66.80±14.33 | **88.00±21.54** | 71.40±28.60 | 71.07±17.15 | 27.52±7.48 | 62.56±37.64 | 67.38±6.58 |
| H13+H18-PC | 3 | 6.20±5.15 | 43.20±45.87 | 47.60±40.66 | 8.56±14.65 | 57.30±11.84 | 31.73±26.33 | **77.45±12.16** |
| H13+H18-PC | 4 | **99.40±0.89** | 32.60±31.15 | 42.00±41.86 | 83.79±17.54 | 53.62±14.30 | 90.38±20.01 | 96.65±1.81 |
| H13+H18-PC | AA(%) | 64.40±3.58** | 41.20±14.91** | 63.60±11.35 | 65.23±4.65** | 56.81±3.67** | 52.43±4.55** | **77.79±1.93** |
| H13+H18-PC | OA(%) | 62.89±2.68** | 45.64±18.33** | 56.96±15.58 | 60.62±6.17** | 55.26±6.88** | 57.87±4.34** | **80.74±3.47** |
| H13+H18-PC | $\kappa$(×100) | 51.00±3.39** | 23.60±23.39** | 0.45±0.18** | 48.80±7.06** | 40.79±7.98** | 42.71±5.47** | **73.29±4.87** |
| PU+PC-H13 | 1 | 13.20±13.23 | 9.00±13.89 | 19.60±20.78 | 55.04±16.84 | 60.35±6.42 | 12.90±15.70 | **78.76±12.68** |
| PU+PC-H13 | 2 | 12.40±9.27 | 51.00±31.76 | 73.40±30.34 | **91.30±13.82** | 21.31±2.75 | 85.70±10.84 | 74.96±5.09 |
| PU+PC-H13 | 3 | 59.80±33.30 | **95.00±3.16** | 62.40±36.24 | 17.44±12.73 | 80.36±2.98 | 86.16±12.53 | 87.00±8.84 |
| PU+PC-H13 | 4 | **99.40±1.14** | 79.60±27.75 | 86.00±9.53 | 76.56±18.00 | 92.71±4.96 | 79.99±39.45 | 95.76±1.48 |
| PU+PC-H13 | AA(%) | 46.00±8.48** | 58.80±5.77 | 59.40±6.67* | 59.90±3.49** | 63.68±2.67** | 66.70±6.89** | **82.91±0.89** |
| PU+PC-H13 | OA(%) | 46.25±8.47** | 58.70±4.86 | 59.37±5.99* | 59.90±3.47** | 63.74±2.69** | 66.63±7.26** | **82.61±0.86** |
| PU+PC-H13 | $\kappa$(×100) | 28.40±10.68** | 45.00±6.32 | 0.46±0.08** | 46.83±4.68** | 51.64±3.33** | 55.53±8.81** | **77.46±1.06** |
| PU+PC-H18 | 1 | 63.40±37.68 | 6.20±4.76 | 0.80±1.30 | 21.49±9.68 | **71.87±13.05** | 71.66±22.05 | 64.99±7.34 |
| PU+PC-H18 | 2 | 49.20±11.97 | 66.00±5.72 | 59.40±15.83 | **84.75±6.63** | 66.98±14.18 | 74.61±13.43 | 63.47±8.34 |
| PU+PC-H18 | 3 | 17.60±18.25 | **99.60±0.55** | 53.60±33.14 | 19.94±3.31 | 85.62±3.08 | 72.80±20.42 | 90.46±6.31 |
| PU+PC-H18 | 4 | **93.60±5.79** | 30.40±19.17 | 57.20±30.01 | 14.87±2.04 | 72.63±2.17 | 32.57±15.19 | 91.99±3.65 |
| PU+PC-H18 | AA(%) | 55.80±10.44** | 50.40±4.92** | 43.00±17.75* | 35.31±1.53** | 74.27±2.78 | 60.13±11.22 | **76.25±1.56** |
| PU+PC-H18 | OA(%) | 72.79±5.26* | 42.10±12.31** | 52.29±22.05* | 21.09±2.15** | 74.58±1.68** | 47.74±12.24** | **85.27±0.86** |
| PU+PC-H18 | $\kappa$(×100) | 46.20±10.91** | 26.40±9.19** | 0.29±0.27** | 7.73±1.06** | 60.74±2.25** | 32.36±13.31** | **73.98±0.69** |

are dynamically extracted, and prototype-based labels for target samples are assigned through cosine similarity-based prototype matching. A customized learning strategy is further introduced by exploiting the consistency between prototype-based labels and predicted pseudo-labels, realized through inter- and intradomain contrastive learning.

For each of the above-mentioned methods, we conducted five independent experiments with fixed random seeds ranging from 0 to 4 to ensure reproducibility and to account for variability caused by random initialization. The results are reported as mean ± standard deviation, which provides a comprehensive reflection of the stability and distribution of model performance across multiple runs. All comparison methods were trained and tested under identical experimental settings, and their corresponding codes were obtained from the official implementations provided by the authors.

In addition, paired $t$-tests were performed to further evaluate the statistical significance of the observed performance differences. Specifically, under the four experimental settings and for the three evaluation metrics (OA, AA, and $\kappa$), pairwise significance tests were conducted between our proposed PADA and each of the comparison methods, based on the results of the five repeated experiments. As shown in Table II, results with $p < 0.05$ are marked with "*" and those with $p < 0.01$ are marked with "**," indicating statistically significant improvements of PADA over the compared methods. This rigorous analysis confirms that the superiority of PADA is not due to random fluctuations but reflects its robustness and reliability in feature learning and cross-domain generalization.

*1) Quantitative Comparison:* To evaluate the effectiveness of our proposed method, we compare it with several state-of-the-art approaches across four challenging cross-domain scenarios, including H13+H18-PU, H13+H18-PC, PU+PC-H13, and PU+PC-H18, where, for example, H13+H18-PU indicates that H13 and H18 are used as the source domains, and PU is used as the target domain. The OA, AA, and $\kappa$ are used as the core evaluation metrics. The best and the second-best results are highlighted in bold and underlined in Table II, respectively.

In the H13+H18-PU task, our method achieves the highest OA of 73.13%, significantly surpassing most baselines ($p < 0.01$). It also obtains the highest AA (76.15%) and $\kappa$ (61.52), with all differences confirmed as statistically significant compared to other methods. Among the four classes, PADA reaches 100.00% in Class 4 and ranks second in Classes 2 and 3, demonstrating robust performance across both large and small categories. The significant results further indicate that the performance gains are not due to random chance.

In the H13+H18-PC setting, the proposed method again achieves the best OA of 80.74%, which is significantly higher than all baselines ($p < 0.01$ in most cases). Our method also obtains the highest AA (77.79%) and $\kappa$ (73.29), showing strong robustness under complex land-cover categories. Notably, in Classes 3 and 4, PADA achieves 77.45% and

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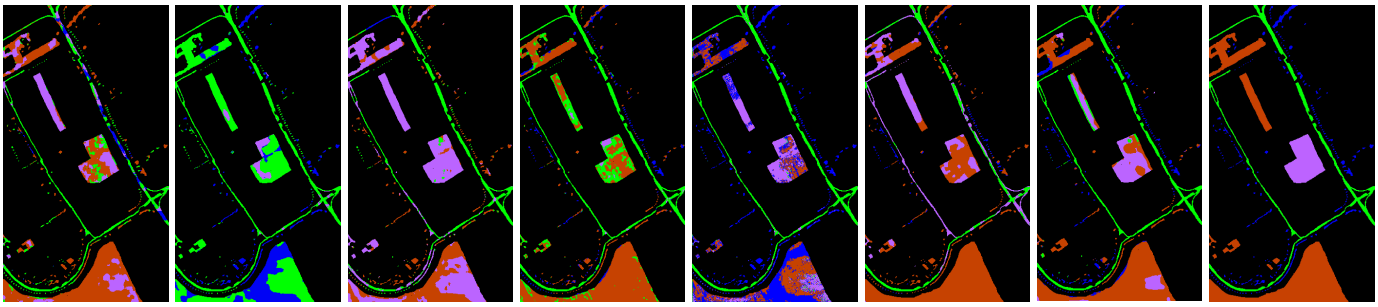



Fig. 4. Visual comparison on the PU dataset. (a) SSFTT. (b) SSTN. (c) AMS-M2ESL. (d) SDEnet. (e) MDGTnet. (f) CACL. (g) PADA (ours). (h) Ground-truth map.

96.65%, respectively, both with significant improvements over competing methods. These results confirm that PADA can effectively reduce domain discrepancies, with improvements validated by statistical tests.

For the PU+PC-H13 task, our method obtains the best OA (82.61%), AA (82.91%), and $\kappa$ (77.46), all significantly higher than competing methods ($p < 0.01$). In particular, PADA surpasses the second-best method CACL by more than 16% in OA and 17% in AA, highlighting its advantage in multisource scenarios with large domain gaps. PADA also delivers top performance in Class 1 (78.76%) and Class 4 (95.76%), further illustrating its strong generalization capability.

In the PU+PC-H18 scenario, which presents greater interdomain variance, our method still maintains the best performance, achieving 85.27% OA, 76.25% AA, and 73.98 $\kappa$. Compared with the second-best method, MDGT, PADA improves OA by 10.69% and $\kappa$ by 13.24, both with $p < 0.01$. Moreover, PADA achieves leading accuracies in Class 3 (90.46%) and Class 4 (91.99%), demonstrating its effectiveness in capturing fine-grained and spatially structured features.

Overall, across all four experimental settings, our method consistently achieves the best results in all three evaluation metrics, with nearly all improvements over the baselines verified as statistically significant. This strong performance can be attributed to the proposed source-and-target shared information mining module and the multisource collaborative classification module, which together enhance the robustness and discriminability of the learned features. The incorporation of statistical significance tests further confirms that the superior performance of PADA is not due to random fluctuations but reflects reliable improvements in cross-domain HSI classification.

*2) Visualization Comparison:* Fig. 4 presents the classification maps of the PU dataset generated by different methods, including SSFTT, SSTN, AMS-M2ESL, SDEnet, MDGTnet, CACL, and the proposed PADA, along with the corresponding ground-truth map. As illustrated in the figure, the proposed PADA method delivers the most accurate and visually consistent classification results, showing a higher resemblance to the ground-truth map compared to the other methods.

Specifically, PADA exhibits clearer region boundaries, reduced salt-and-pepper noise, and better preservation of small-scale structures such as roads and trees. Notably, in the central region of the image, which corresponds to the bare soil class, PADA delivers significantly more accurate and spatially coherent predictions. Methods like the SSFTT and SDEnet tend to produce fragmented and noisy results in this area by misclassifying large portions of bare soil into surrounding categories. The performance of SSTN is even less satisfactory, as it correctly classifies only a few pixels in this region. In contrast, PADA effectively delineates the boundaries of the bare soil class and preserves its structural integrity, demonstrating superior spatial discrimination and robustness in handling challenging category regions.

In contrast, other methods, such as the MDGTnet and AMS-M2SEL, suffer from class confusion and often fail to identify certain regions correctly—particularly in the case of elongated structures or fine-grained boundaries. CACL performs relatively well but still introduces some classification errors in small object areas.

Overall, the visual results confirm that the proposed PADA model can generate cleaner and more accurate classification maps, effectively mitigating domain shifts and enhancing fine-grained recognition in challenging multisource scenarios.

### D. Ablation Studies

To evaluate the effectiveness of key components in our model for cross-dataset HSI classification, we conduct ablation studies by comparing the original PADA with its following variants.

*1) Effectiveness of the Source-and-Target Shared Information Mining Module:* To evaluate the impact of single-source versus multisource-domain adaptation, we simplify the $N$ parallel shared information mining branches into a single-branch structure, effectively reducing it to a single-source domain adaptation framework. As shown in Table III, using only one source HSI results in decreased accuracy and consistency compared to employing two source HSIs. This highlights the effectiveness of our source-and-target shared information mining module in aligning multisource HSI features.

*2) Effectiveness of the Global Context-Aware Spectral Feature Extractor:* To quantify the contribution of global spectral modeling to feature representation, we remove the global context-aware spectral feature extractor while retaining only the local structure-aware spatial feature extractor. As shown

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TABLE III
CLASSIFICATION RESULTS OF ABLATION STUDIES IN SINGLE-SOURCE AND MULTISOURCE HSIS

| Class | SD-TD | | |
|---|---|---|---|
| | PU-H18 | PC-H18 | PU+PC-H18 |
| 1 | 71.80 | **80.60** | 76.23 |
| 2 | **80.46** | 20.88 | 65.69 |
| 3 | 41.62 | **88.47** | 72.63 |
| 4 | 92.95 | 86.36 | **93.06** |
| AA(%) | 72.31 | 69.00 | **76.40** |
| OA(%) | 80.18 | 81.49 | **85.12** |
| $\kappa(\times 100)$ | 65.41 | 68.36 | **73.47** |

TABLE IV
CLASSIFICATION RESULTS OF ABLATION STUDIES IN THE GLOBAL CONTEXT-AWARE SPECTRAL FEATURE EXTRACTOR AND THE SOURCE-TO-TARGET TRANSFERABILITY LEARNING MODULE

| Class | PU+PC-H18 (SD-TD) | | |
|---|---|---|---|
| | w/o Spectral Extractor | w/o Domain Consensus | PADA |
| 1 | **82.00** | 67.78 | 74.23 |
| 2 | **94.71** | 92.18 | 65.69 |
| 3 | 23.65 | 50.09 | **72.63** |
| 4 | 87.11 | 74.02 | **93.06** |
| AA(%) | 71.86 | 71.02 | **76.40** |
| OA(%) | 75.20 | 69.88 | **85.12** |
| $\kappa(\times 100)$ | 59.76 | 61.87 | **73.47** |

in Table IV, relying solely on TriConvBlock for feature extraction leads to a significant decline in OA, indicating that the spectral–spatial feature extractor effectively captures discriminative feature representations.

*3) Effectiveness of Source-to-Target Transferability Learning Module:* To assess the impact of dynamic weight allocation on heterogeneous feature fusion, we remove the source-to-target transferability learning module and replace the correlation-aware fusion mechanism with a simple averaging strategy. As shown in Table IV, the model's classification performance deteriorates without the source-to-target transferability learning module, validating the effectiveness of the source-to-target transferability learning module.

## V. CONCLUSION

In this article, we propose PADA, a novel and efficient framework for cross-dataset HSI classification. The framework improves computational efficiency through a spectral–spatial feature extractor and overcomes the scalability limitations of traditional single-source domain adaptation by introducing a source-and-target shared information mining module. Moreover, a correlation-aware fusion mechanism is designed to adaptively integrate knowledge from multiple source HSIs, enhancing both feature alignment and target classification accuracy. Extensive experiments on four widely used benchmark datasets demonstrate the superior performance of PADA over state-of-the-art methods, offering a new perspective on efficient and scalable domain adaptation for cross-dataset remote sensing data.

Despite these advancements, several promising research directions remain. We plan to explore self-supervised and contrastive learning paradigms to enhance feature robustness, particularly in scenarios with limited labeled hyperspectral data, leveraging spectral similarity constraints or contrastive embeddings to improve cross-domain generalization. We also aim to design lightweight variants of PADA for deployment on edge devices and to extend the framework to unseen geographic regions and novel land-cover types, thereby enhancing practical applicability, scalability, and robustness in real-world remote sensing scenarios.

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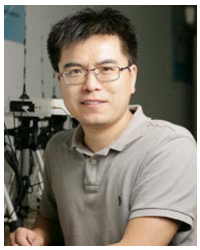
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